



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

Program/s: BTech

Year: II Semester: IV

Stream/s ;Data Science

Subject: Stochastic processes and applications

Time: 03 hrs (10:00am to 1:00pm)

Date: 02 / 05 / 2023

No. of Pages: 03

Marks: 100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.
- 8) It is open book/open notes exam.

Q1		Answer briefly:	[20]
	a.	Random variable x has CDF $F_x(x) = \begin{cases} 0 & x < 0, \\ x/2 & 0 \leq x \leq 2, \\ 1 & x > 2, \end{cases}$ 1- What is $E[X]$? 2- What is $Var[X]$?	[5]
	b.	The number of costumers K that arrive at the restaurant in one hour is a Poisson random variable with $E[K] = 8$. Find $P[K = 7]$.	[5]
	c.	Sketch the transition diagrams of the following Markov chain $A = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$	[5]
	d.	Consider a Markov chain with state space $\{0,1,2\}$ and	[5]

		transition matrix $P = \begin{bmatrix} 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \end{bmatrix}$ 1- Is the chain reducible? Justify your answer. 2- Is the chain periodic? Justify your answer. 3- Does the chain have transient states? Justify your answer.	
Q2	a	Random variables X and Y have joint PDF $f_{X,Y}(x,y) = \begin{cases} cx & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$ a- Find the constant c. b- Find the marginal PDF $f_X(x)$. c- Are X and Y independent? Justify your answer.	[20]
Q3	a	Determine whether each of these statements is true or false. Justify your answer for full credit. (a) If X(t) and Y (t) are independent wide sense stationary processes, then the process $W(t) = X(t)Y(t)$. Is a wide-sense stationary: (b) Given a Gaussian process X(t), then $2X(t) - 3X(t - 1)$ is not a Gaussian process.	[20]
Q4	a	A production capacity C(n) at a given factory can be given by $C_n = \frac{C_0}{2^n} + 4 \sum_{i=1}^n \frac{X_i}{2^{n-1}}$ where $C_0, X_1, X_2, \dots$, is an iid random sequence of Gaussian (0,1) random variables. a) Find the mean of C_n. b) Is C_n a Gaussian random variable? c) Is C_0 a normal random variable, if yes why?	[20]
Q5	a	Consider a Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ with state space $S = \{0, 1\}$ and one-step transition probability $P = \begin{bmatrix} 0.5 & 0.5 \\ 1 & 0 \end{bmatrix}$ a- Is the chain irreducible? b- Is the chain periodic?	[20]

		c- Is the chain positive recurrent? d- Find the limiting probabilities.	
Q6	a	Consider a distribution center that collects packages arriving according to exponential distributed inter-arrival times. Assume that the delivery time of a package is exponential distributed with the mean value of 10 min. It is known that the idle state probability of the distribution center is 0.8. a- What type of a queue model can be used to describe the work of the distribution system? b- Determine the service rate μ and arrival rate λ. c- Determine the mean delay (waiting time) for the package.	[20]